Business Summary Report: Predictive Insights for Collections Strategy

Prepared for the Head of Collections, Geldium | Helga Yuliani Putri Aritonang, AI Transformation Consultant, Tata iQ | October 2026

Bottom line: Today's data cannot yet reliably predict which individual customers will miss payments; our model currently performs no better than guessing, so it should not be used to decide how any single customer is treated. Three customer patterns do point to higher risk, and they are strong enough to guide a **low-cost, supportive outreach pilot** with a control group, while the data is fixed in parallel.

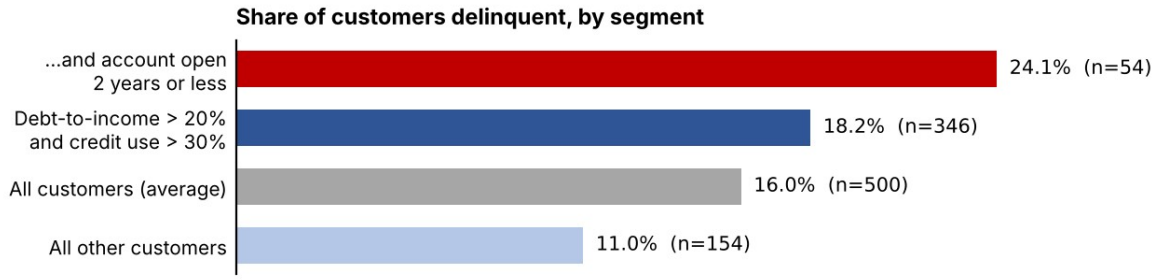
1. Summary of Predictive Insights

Of 500 customers reviewed, 80 (16%) are delinquent. The three factors below were linked to higher delinquency. The differences are modest and should be confirmed by testing before being used at scale.

Top 3 Risk Factors

- **High debt relative to income** (Debt_to_Income_Ratio): customers whose debt exceeds 20% of income were delinquent 17.2% of the time, vs 9.8% for those at or below 20%.
- **Heavy use of available credit** (Credit_Utilization): customers using more than 30% of their limit were delinquent 16.8% of the time, vs 12.0% for lighter users; 4 customers are already over their limit.
- **New customer relationships** (Account_Tenure): customers with Geldium for 2 years or less were delinquent 20.8% of the time, vs 15.1% for longer-standing customers.

Highest-risk segment: customers with **both** high debt-to-income and credit use above 30% (346 customers) account for **63 of the 80 delinquent customers (79%)**. Within them, the 54 customers with accounts open 2 years or less have the highest rate, at 24.1%.



What the team should know: payment history (Missed_Payments, Month_1–6) is normally the strongest warning sign, but in this data it conflicts with itself for 141 customers (28%), so it cannot be relied on until corrected. Age and location also showed differences, but we do not recommend using them (see Section 3).

Key Insight	Customer Segment	Influencing Variables	Potential Impact
High debt plus heavy credit use concentrates risk	346 customers; 79% of all delinquent cases	Debt_to_Income_Ratio > 20%, Credit_Utilization > 30%	Focus outreach on about 70% of the book while still reaching most at-risk customers
Newer accounts are riskiest within that group	54 customers; 24.1% delinquent	Account_Tenure ≤ 2 years	Contact this group first within the pilot
Current model is not yet reliable	All customers	Conflicting payment-history fields	Fix data before any automated, model-based decisions

2. Recommendation Framework

- **Restated insight:** Customers whose debt exceeds 20% of income and who use more than 30% of their credit limit make up 79% of delinquent accounts, but the pattern is modest and has not yet been proven to respond to intervention.
- **Proposed recommendation:** Run an 8-week supportive outreach pilot for this segment, with half the customers contacted and half kept as a comparison group, to test whether early, helpful contact reduces missed payments.
- **Specific:** Randomly split the 346-customer segment. The outreach group receives an SMS and email reminder 5 days before each due date, plus an offer to discuss a flexible repayment plan; the comparison group receives standard treatment. Customers with accounts open 2 years or less are contacted first.
- **Measurable:** Reduce the 30+ day delinquency rate in the outreach group by at least 10% relative to the comparison group, while tracking opt-outs and complaints.
- **Actionable:** Uses existing SMS, email and agent teams; selection uses two fields already held, so no new model is needed.

- **Relevant:** Targets the segment holding most delinquent accounts, with support that addresses stretched budgets (the core issue behind high debt-to-income and credit use).
- **Time-bound:** Launch within 2 weeks; run for 8 weeks (two billing cycles); report results to the Head of Collections in week 10.
- **Justification and business rationale:** The pilot is cheap, low-risk to customers, and produces hard evidence of what works. Because results are measured against a comparison group, they do not depend on the model being accurate. In parallel, the analytics team will correct the payment-history and delinquency fields within 4 weeks and re-test the model. Geldium should scale outreach only if the pilot meets its target, and use the model only once it clearly beats chance.

> **For stakeholders:** We recommend testing gentle, early reminders and repayment support on the customers carrying the most debt, because this group holds 79% of today's delinquent accounts. The test will show within 10 weeks whether the approach reduces missed payments before Geldium invests more.

3. Ethical and Responsible AI Considerations

Two fairness risks and how we mitigate them

- **Indirect discrimination through age and location.** Customers aged 46–60 and some cities showed higher delinquency, and card types such as "Student" may stand in for age. Mitigation: exclude age, location and age-linked fields from targeting, and compare contact and outcome rates across age bands and cities; any group treated at less than 80% of the rate of another triggers a review.
- **Customers wrongly labelled as high risk because of bad data.** With 28% of payment records in conflict and a model that is not yet reliable, some customers could be mislabelled. Mitigation: use scores only for supportive, opt-out contact (never penalties or credit-limit cuts), keep a collections officer in the loop for any action affecting an account, and correct the data before any automated use.

Explainability and responsible use

- **Plain-language reasons:** every contacted customer can be told why ("your balance is high relative to your income and credit limit"); the selection rule is two clear criteria, not a black box.
- **Transparency and accountability:** the Head of Collections owns pilot decisions; results, complaints and fairness checks are reviewed at weeks 4 and 10.
- **Data privacy:** outreach uses only the fields needed for selection, and customer identifiers stay inside Geldium systems.